DDM501 · INDIVIDUAL ASSIGNMENT 1 · ML SYSTEM DESIGN DOCUMENT

# Early Warning System for Students at Risk of Dropping Out


| | |
|---|---|
| Student | **Nguyen Thi Hong Hanh** |
| Student ID | **25MS13316** |
| Domain | Higher (college) education |
| Problem type | Binary classification with risk ranking (weekly batch scoring) |
| Case context | BTEC FPT British College (all figures are assumed for illustration) |


> Note on data: the institution provides the context of the case, but student numbers, dropout rates, tuition fees and staff numbers are **assumptions** made for this design exercise and do not represent official figures.

## Executive Summary


BTEC FPT British College (assumed: 12,000 students) loses about **8% of its students every year** to dropout. The college already has a student care team that calls students when they miss classes, but this mechanism reacts to individual absences and looks at only one signal; formal academic warnings arrive only after module results are published, which is usually too late to help.

This document designs a machine learning system that **scores every enrolled student's dropout risk every week** and splits outreach into two tiers: **academic advisors** receive a short high-priority list with reasons so they can meet and counsel students, and the **student care team** receives a follow-up call list. In this way both existing human resources are directed to the right students early.

The system runs as a weekly batch job using a calibrated gradient boosting model, compared against the current academic warning rule and a logistic regression baseline. The business goal is to reduce the annual dropout rate from 8% to 6.5% within two academic years. Final decisions always remain with people: the system **suggests** and never sends risk warnings to students automatically.


## 1. Problem Definition

Part 1 · 20%

### 1.1 Context

BTEC FPT British College delivers BTEC programmes to an assumed population of about **12,000 students**. Students **study module by module**, with **three main semesters per year**, each lasting **15 weeks**. The college has **40 academic advisors**, each responsible for roughly 300 students in addition to their teaching duties. The college also has **lecturers** who run each module (attendance, assessment) and a dedicated **student care team** that phones students to remind them when they miss classes.

Each year about **960 students (8%)** drop out: they do not register for any module in the next semester, formally withdraw, or are dismissed. With three semesters per year, students have **three exit points** annually, so on average about 320 students are lost each semester. First-year students have the highest rate. The impact has three dimensions:

- **College finances:** with an assumed average tuition fee of VND 25 million per year (to be replaced with the actual fee), one lost year for 960 students equals about **VND 24 billion** in revenue.

- **Reputation and quality assurance:** retention and on-time completion are indicators considered in quality assurance and in the school choice of students and parents.
- **Students and families:** lost time, money already spent and career opportunities. Many dropouts are caused by difficulties that can be addressed (finances, study skills, wellbeing) if detected early.

## 1.2 Problem Statement

> PROBLEM
> Every week during the semester, identify students at high risk of **not returning in the next semester**, at least **8 weeks** before the semester ends, so that the **40 academic advisors** (capacity for in-depth counselling of about **10% of students per semester**, i.e. about 30 students each) and the **student care team** (follow-up calls to roughly the next **15% of students**) can contact the right students first.
>
> MEASURABLE OBJECTIVE
> Reduce the annual dropout rate from **8% to 6.5%** within two academic years, i.e. retain about **180 additional students per year**.

**Label definition** (critical because it determines the training data): a student enrolled in semester *t* is labelled 1 if they **register for no module in semester t+1** and are not graduating, on approved leave, or transferring with documentation; or if they are dismissed. The label is only certain after the registration deadline of semester t+1, so **labels arrive about 2–4 weeks** after the semester ends (faster than in a two-semester calendar because semesters follow each other closely). The positive rate is only about **2.7% per semester** (≈ 8% per year divided across three semesters), so the data is highly imbalanced.

## 1.3 Current Situation

- **Formal academic warnings:** the Academic Office issues a warning when a student fails modules or falls below the required standard. Warnings arrive only **after a module ends** and consider only grades.
- **Student care calls on absence:** an existing strength, but the current practice is **reactive, absence by absence** (a hard rule such as "call after N absences"). Every absence is treated the same, callers do not know whether the student also has unpaid fees or declining grades, and call outcomes are not passed to advisors in a structured way. Students who attend regularly but stop submitting work or fall behind on fees are not called.
- **Manual monitoring by advisors:** advisors review transcripts and lecturer feedback when time allows. With 300 students each plus teaching, this cannot be done consistently.
- **Fragmented data** across the student information system (results, module registration), the LMS, attendance, tuition accounting and the call log. Nobody sees the combined picture.

Consequence: the college **has people to contact students but lacks the information to decide whom to contact first**. Most cases are only discovered when a student fails to register for the new semester. Students whose grades are acceptable but who leave because of **unpaid fees** or **loss of motivation** (visible as absences and LMS inactivity) are almost never detected.

## 1.4 Why Machine Learning Is Appropriate

| Screening question | Answer |
|---|---|
| Can success be defined and measured? | **Yes.** The label is clear (registered next semester or not); dropout rate, recall and precision are measurable. |

| Screening question | Answer |
|---|---|
| Is there enough data? | **Yes.** Five years of history in existing systems: about 180,000 student-semester records (12,000 × 3 semesters × 5 years), including about 4,800 dropouts. |
| Is the problem too complex for simple rules? | **Yes.** Risk emerges from combinations of signals: declining results, rising absence, LMS inactivity and unpaid fees. A "grades below threshold" rule misses non-academic cases and comes too late. |
| Are mistakes acceptable? | **Yes, if designed correctly.** The consequence of a prediction is *a supportive conversation*, not a penalty. Contacting a student who is not at risk costs about 30 minutes of an advisor's time or a 5–10 minute call from the care team. This is why risk scores are never used for adverse decisions (FR6). |

## 1.5 Stakeholders

| Stakeholder | Goals | Concerns | Needs from the system |
|---|---|---|---|
| **Board of management** | Reduce dropout, protect revenue, meet quality indicators | Investment cost; reputational risk if data leaks | Effectiveness report every semester |
| **Academic advisors** (primary users) | Support the right students with limited time | Too many false alerts; extra data entry | Short list with reasons; few clicks |
| **Student care team** (secondary users) | Call the right students; stay in touch with those showing signs of leaving; coordinate support | Call lists too long; duplicate calls with advisors; unclear what to say | Weekly call list with short reasons; quick logging of call outcomes; escalation to advisors; aggregate dashboard by programme and cohort |
| **Lecturers** | Students keep up with modules | Additional reporting work | No extra work: reuse attendance and LMS data; record attendance on time |
| **Students** | Receive help when struggling | Being labelled, monitored, treated differently; exposure of financial information | Transparency about data use; risk scores never affect grades or scholarships |
| **Data and IT team** | Stable, maintainable system | Source data changes, model degradation | Automated pipelines, monitoring, rollback |
| **Data protection / legal officer** | Compliance with personal data protection rules | Use of student data beyond its purpose | Access control, audit logs, data minimisation |

**Key conflicts** (revisited in Part 5):

- Management and the data team want **more data** for better predictions; students and legal want **as little data as possible**.
- Management wants to **catch as many cases as possible** (high recall); advisors want a **short, precise list** (high precision, within capacity). The resolution is two tiers: a short list for advisors and a longer list for the care team, whose cost per contact is low.
- Management wants **automation** (mass warning emails); students and advisors want **human contact** and to avoid being labelled.
- Advisors and the care team both contact students; without coordination **one student may be called repeatedly** while another is missed.

# 2. Requirements Analysis

Part 2 · 20%

## 2.1 Functional Requirements

| ID | Requirement |
|---|---|
| **FR1** | **Weekly risk scoring** for every enrolled student: calibrated probability of not returning next semester, from week 3 to week 13 of the semester. |
| **FR2** | **Tiering and assignment:** each week students are split into **Tier A** (top 10% risk college-wide), assigned to their **academic advisor** for a meeting or counselling, and **Tier B** (top 10–25%), assigned to the **student care team** for a check-in call. Each student has **exactly one contact owner** per week to avoid duplicate calls. Advisors see their own students ranked by risk; the care team sees a shared call list. |
| **FR3** | **Prediction reasons:** each listed student comes with the 3 strongest contributing factors (e.g. "missed 40% of classes in the last 4 weeks", "no LMS login for 12 days", "tuition overdue 45 days") so the contact person can choose an appropriate approach (e.g. ask about the timetable when absence is the reason, refer to the finance office when fees are the reason). |
| **FR4** | **Intervention logging and escalation:** advisors and care staff record that contact was made, the channel (meeting, phone, referral to counselling, fee support) and the outcome. Care staff can **escalate a case to the advisor** when a call reveals a problem requiring in-depth counselling. This data is used for evaluation and retraining. |
| **FR5** | **Aggregate dashboard** for the head of student care and the board: risk distribution by programme and cohort, contact rates, weekly trends. No individual risk scores at this level. |
| **FR6** | **Usage restrictions:** risk scores are **not** shown to students or module lecturers and are not used for scholarships, discipline or admissions. The system never sends notifications to students by itself. |

### Input and output specification

| Group | Inputs (computed as of week w of the semester) | Source |
|---|---|---|
| Academic | Results of completed modules (converted average, cumulative failed modules, failed modules last semester), trend over the last two semesters, number of modules registered this semester, modules being retaken | Student information system |
| Attendance and care | Absence rate in the last 4 weeks and since semester start, number of modules with >20% absence, number of care-team reminder calls and whether the student was reachable | Attendance system; care-team call log |
| Engagement | LMS login days per week, days since last login, on-time submission rate, available interim assessment results | LMS |
| Financial | Outstanding tuition, days overdue, scholarship or fee reduction | Tuition accounting |
| Study context | Number of semesters completed, programme, admission route, programme change, current week of semester | Student information system |

**Excluded from the model:** gender, ethnicity, religion, hometown, health information, and the content of student messages or posts. Gender and region are kept in a separate store only for **fairness auditing** (Section 3.4).

**Weekly output per student:** `student_id` (pseudonymised in the analytics store), `risk_score` (0–1, calibrated), `tier` (A / B / none), `top_factors` (3 factors with direction), `contact_owner`, `model_version`, `scored_week`.

### Integration with existing systems

- **Read data** from the source systems via read-only replicas or nightly exports; never write back to source systems.
- **Single sign-on (SSO)** with staff accounts; advisors see only their assigned students, care staff see only the weekly call list.
- **Notifications:** every Monday morning, advisors receive an email with the number of new Tier A students and the care team receives the week's call count, each with a link to the list (emails contain no personal data).
- **Coordination with the existing reminder process:** same-day absence calls by the care team **continue unchanged**; the system adds a weekly prioritised list and reads the call log to avoid duplicate contact.

### User interaction requirements

On the intervention scale (annotate → ask → suggest → automate → enforce), the system operates at the **suggest** level: advisors and care staff see the list and the reasons, and decide whether and how to contact each student. Rationale: model confidence is moderate, and an automated warning sent directly to a student may cause distress. Contact persons can mark "no contact needed" with a reason; this feedback is used to improve the model.

## 2.2 Non-Functional Requirements

| ID | Category | Requirement |
|---|---|---|
| **NFR1** | Performance | Weekly batch scoring starts at 02:00 Sunday and **completes before 06:00 Monday** (scoring 12,000 students is expected to take under 5 minutes; the remaining time covers extraction and validation). The advisor list page loads with **P95 < 2 seconds**. |
| **NFR2** | Scalability | Serve 12,000 students, 40 advisors, about 10 care staff (assumed) and about 30 managers. Designed for **50,000 students** (growth or multiple campuses) without architectural change: batch scoring scales linearly and can be partitioned by programme. |
| **NFR3** | Reliability | Dashboard availability **99% during office hours**. If weekly scoring fails: keep last week's list, show its date, alert the data team. If source data validation fails: do **not** score on bad data; switch to a **fallback rule** (absence >30% or tuition overdue >30 days or ≥2 failed modules last semester). |
| **NFR4** | Maintainability | Retrain **every semester** (three times a year) once labels are complete, or earlier when drift is detected (PSI > 0.2 on key features). A new model is used only after passing the quality gate and human approval. All versions are stored for rollback within 5 minutes. |
| **NFR5** | Security and audit | Role-based access; encryption in transit and at rest; every view of an individual list is logged and retained for 12 months. |

**Why batch instead of real time:** source data (attendance, LMS, fees) is updated daily, advisor outreach happens on a weekly rhythm, and same-day absence calls already have their own process. Minute-level risk updates would not change any decision while adding significant infrastructure (see Part 5).

## 2.3 Data Requirements

| Aspect | Requirement |
|---|---|
| **Sources** | The source systems in Section 2.1, extracted nightly, plus the care-team call log and interventions recorded by advisors (FR4). |

| Aspect | Requirement |
|---|---|
| **Quality** | **Accuracy:** attendance may be missing when lecturers forget to record it, so absence rates use only sessions with data and carry a "missing attendance" flag.<br>**Completeness:** mandatory fields (student ID, semester, registered modules) must be 100% complete; LMS data may be missing for modules that do not use the LMS, handled with missing-indicator features.<br>**Consistency:** standardise student and semester codes across systems; check referential integrity.<br>**Timeliness:** data must be updated within 48 hours before scoring, otherwise the fallback rule is used (NFR3).<br>Every extraction is checked for schema, value ranges and distributions before use. |
| **Privacy** | Student data is personal data; financial information requires special protection. Comply with current Vietnamese personal data protection regulation (Decree 13/2023/ND-CP and any successor). Measures: inform students that data is used for academic support; **pseudonymise** the analytics store (hashed IDs, mapping table stored separately); only the assigned contact person sees identities; no secondary use (FR6); delete detailed LMS logs after 2 years and keep only aggregated features. |
| **Volume** | Five years × three semesters ≈ **180,000 student-semester records**. LMS logs ≈ 50 million events per year (≈ 20 GB raw), aggregated into weekly features (≈ 12,000 × 15 weeks = 180,000 rows per semester). This fits comfortably on a single server; no big-data infrastructure is required. |

# 3. Goals and Success Metrics

Part 3 · 20%

## 3.1 Goal Hierarchy

| Level | Goal | Metric | Target |
|---|---|---|---|
| **Organisational** | Reduce dropout | Annual dropout rate | 8% → 6.5% in 2 years |
| | Protect tuition revenue | Retained students × annual fee | ≈ 180 students ≈ VND 4.5 bn/year |
| **Leading indicators** | Advisors act promptly | % of Tier A contacted by advisor within 7 days | ≥ 80% |
| | Care team covers Tier B | % of Tier B called within 7 days; % reached | ≥ 90%; ≥ 70% |
| | Students take up support | % of contacted students attending at least one support session | ≥ 50% |
| **System** | Lists on time | % of weeks with lists ready before 06:00 Monday | ≥ 95% |
| | Usable dashboard | Office-hours availability; P95 page load | ≥ 99%; < 2 s |
| | Adoption | % of advisors opening the list weekly; % duplicate calls | ≥ 75%; < 5% |
| **Model** | Good ranking | PR-AUC (time-based test set) | ≥ 0.25 (random ≈ 0.027) |
| | Tier A catches many cases | Recall@10% | ≥ 50% |
| | Tiers A+B catch most cases | Recall@25% | ≥ 75% |
| | Tier A list precise enough | Precision@10% | ≥ 13% (~5× base rate) |
| | Early detection | Recall@10% at week 5 | ≥ 35% |
| | Trustworthy probabilities | Brier score, calibration curve | Better than baseline; calibration error < 0.05 in every score bin |

**How the levels connect:** good ranking (high Recall@10% and Recall@25%) → Tiers A and B contain many truly at-risk students → advisors and the care team contact them promptly → students receive support → some of them stay → dropout falls. The weakest link is not the model but the **intervention success rate**: even a perfect model cannot reduce dropout if nobody makes contact or the support is ineffective. This is why leading indicators are tracked alongside model metrics.

**Sanity check of the business target:** about 960 dropouts per year (≈ 320 per semester). Tier A catches 50% (≈ 480 cases/year); if advisors retain 30% of them, about 144 students stay. Tier B catches a further 25% (≈ 240 cases/year); a lighter check-in call is assumed to retain 15%, about 36 students. The total of about **180 students** matches the 6.5% target. The 30% and 15% rates are **assumptions** to be verified with a control group (Section 3.3).

## 3.2 Metric Rationale

- **No accuracy:** the semester dropout rate is only about 2.7%; a model predicting "nobody drops out" scores about 97% accuracy and is useless.

- **PR-AUC** measures how well at-risk students are ranked at the top, focusing on the minority class. It is used for **model selection**.
- **Recall@10%, Precision@10% and Recall@25%** reflect the operating point: the real constraint is the **contact capacity** of two groups (advisors about 10%, the care team a further 15% of students per semester), so lists are cut by top-k rather than a fixed probability threshold.
- **Recall over precision:** missing a student who will drop out (FN) costs a year of tuition and possibly the student's education; a false alarm (FP) costs about 30 minutes of advisor time or a few minutes of a call. FN cost is many times higher, so the system uses the full contact capacity rather than contacting only very certain cases.
- **Brier score and calibration:** the dashboard displays risk levels, and the care team uses summed probabilities to estimate expected dropouts per programme and allocate staff. These numbers are only meaningful if probabilities match reality.

## 3.3 Baselines and Evaluation

| Baseline | Description | Role |
|---|---|---|
| **B0. Random** | PR-AUC equals the base rate (≈ 0.027) | Lower bound |
| **B1. Current rules** | Academic warning after failed modules, available only after a module ends; plus the "N absences → call" rule | Current practice. The model must **catch more cases, earlier** |
| **B2. Extended rules** | ≥2 failed modules last semester or absence >30% or tuition overdue >30 days | Non-ML alternative; also the fallback rule (NFR3) |
| **B3. Logistic regression** | Simple, interpretable model with the same features | A complex model is chosen only if it beats B3 meaningfully |

**Data split: time-based**, not random: train on older semesters, validate on the following semester, test on the most recent labelled semester. Features for each training record use only data **available at week w** of that semester (point-in-time) to prevent leakage, e.g. final results are never used when simulating week 5.

**Quality gate for a new model:** (1) PR-AUC ≥ 0.25 on the test set; (2) not worse than B3; (3) not worse than the production model by more than 0.01; (4) recall gap between groups (Section 3.4) not above 10 percentage points. Passing the gate still requires human approval.

**Measuring real impact:** model metrics cannot answer "does the system reduce dropout?". In the first semester of deployment, keep a **small random control group** (e.g. 10% of Tier A and B students handled with the existing process, i.e. still receiving absence reminder calls) and compare retention. Because this concerns students' access to support, it requires board approval; the alternative is a staggered rollout by programme with before–after comparison. Three semesters per year allow faster evaluation.

## 3.4 Fairness

Measure **Recall@10% and Recall@25% per group** (gender, region, admission route, semesters completed): among students who actually drop out, every group should have a comparable chance of being detected. Alert threshold: a gap above 10 percentage points. Because being contacted is **support**, missing a group harms that group; a group being contacted more often is a problem only if it feels like labelling, so advisors and care staff follow a common contact guideline (check-in script; never say "you are at risk of dropping out").

# 4. High-Level Architecture Design

Part 4 · 25%

## 4.1 System Architecture Diagram

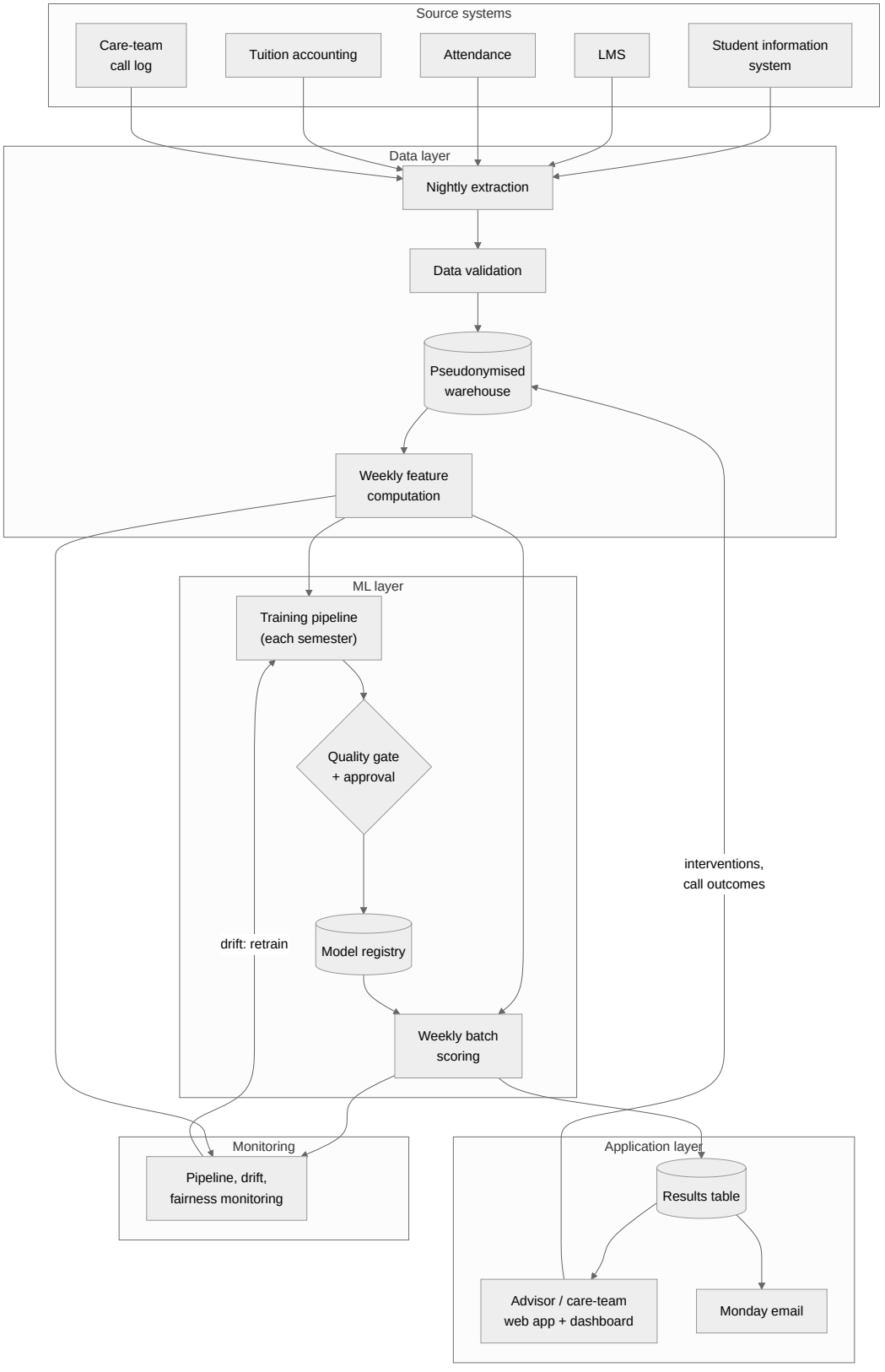



Figure 1. Overall architecture. Scoring runs weekly; training runs each semester or when monitoring detects drift; advisor interventions and care-team call outcomes flow back to the warehouse for evaluation and retraining.

## 4.2 Data Flow and Pipeline Stages

The system has two separate pipelines that share one feature module, which prevents training-serving skew.

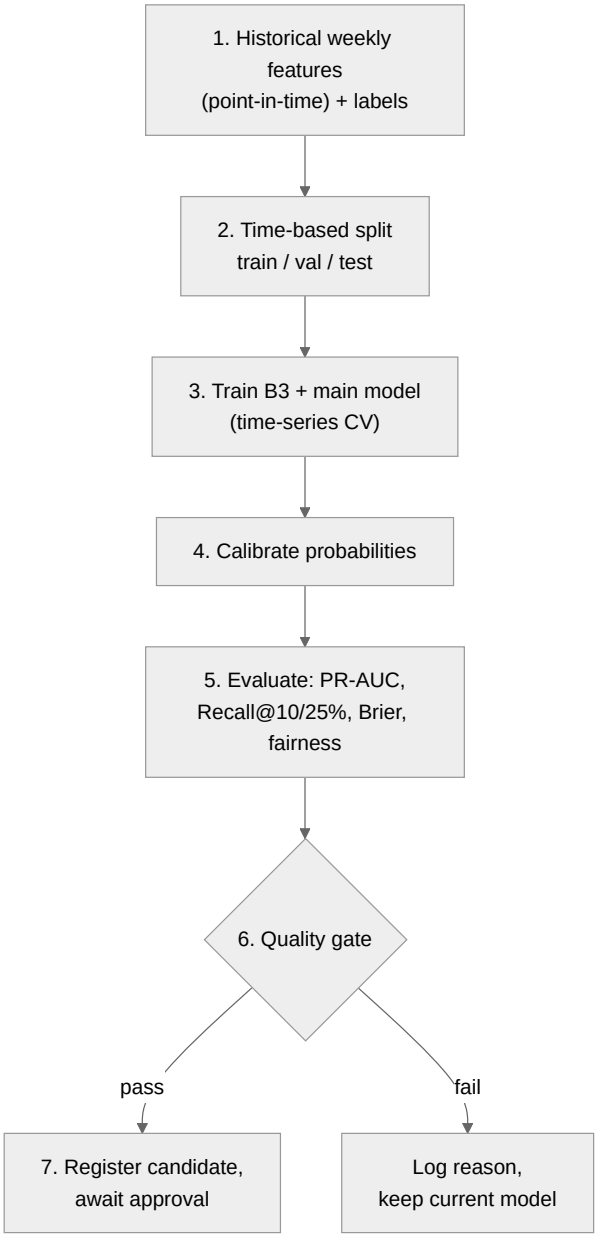



Figure 2. Training pipeline (each semester or on drift).

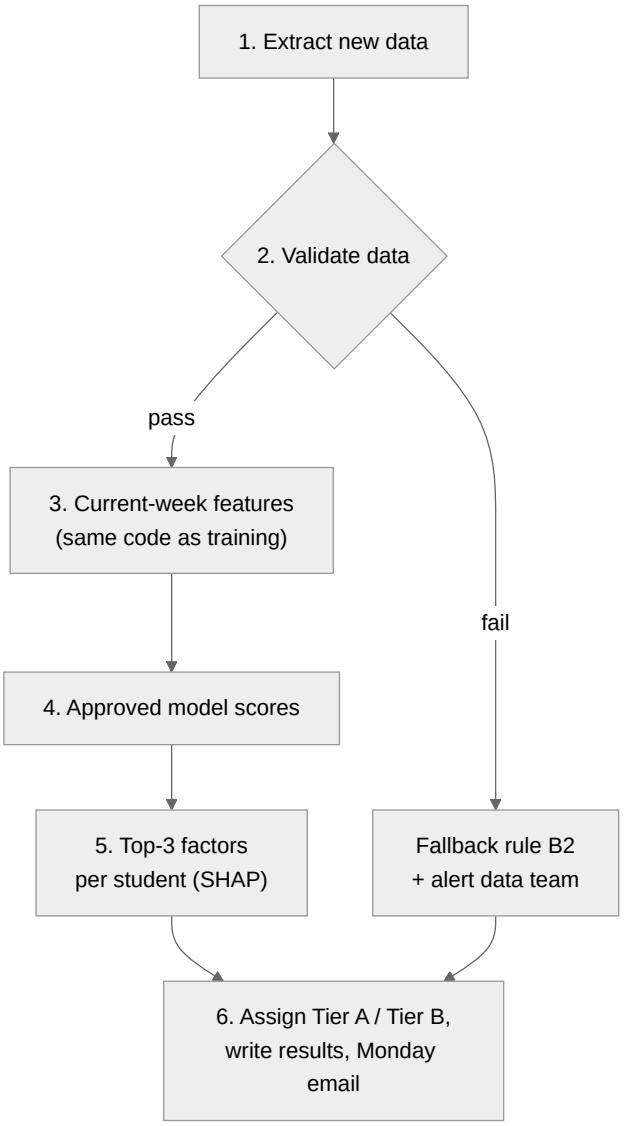



Figure 3. Weekly scoring pipeline.

## 4.3 Component Descriptions

| Component | Purpose and responsibilities | Technology | Rationale |
|---|---|---|---|
| **Orchestration** | Schedule and run extraction, validation, feature computation and scoring; retries; failure alerts | Apache Airflow (lighter option: cron + scripts) | Shows step dependencies and run history; light enough for this scale |
| **Data extraction** | Pull changed data from source systems (including the care-team call log); never write back | SQL on read-only replicas; Python | No load on operational systems |
| **Data validation** | Check schema, ranges, missing rates, distributions; block bad data | Great Expectations or Pandera | Checks defined as code, run inside the pipeline |

| Component | Purpose and responsibilities | Technology | Rationale |
|---|---|---|---|
| **Data warehouse** | Store pseudonymised data, weekly features, labels and interventions | PostgreSQL | Hundreds of thousands of rows; no big-data warehouse needed; familiar to IT |
| **Feature computation** | Turn raw data into point-in-time weekly features | Shared Python module for training and scoring | Defined once, used twice: prevents training-serving skew. No feature store yet: one model, batch only |
| **Training pipeline** | Train, tune, calibrate, evaluate, apply the quality gate | scikit-learn (LR), LightGBM or XGBoost, isotonic calibration | Gradient boosting is strong on tabular data and runs on CPU; LR as interpretable baseline |
| **Experiment tracking and registry** | Record every run (parameters, metrics, data version); manage versions (candidate → production); rollback | MLflow Tracking and Model Registry | Free, self-hosted; aliases separate candidate and production |
| **Batch scoring** | Score all enrolled students, compute top 3 factors with SHAP, assign tiers and contact owners | Python job in Docker, triggered by Airflow | Meets NFR1; SHAP gives per-student explanations |
| **Web app and dashboard** | Tier A lists per advisor, Tier B call list for the care team, reasons, intervention logging and escalation; aggregate dashboard for management | Small web app (FastAPI + simple UI); Metabase or Superset for dashboards | Advisors and care staff need data entry and escalation (FR4); dashboards reuse existing tools to reduce cost |
| **Monitoring** | Pipeline health; feature and score drift (PSI); model performance when labels arrive; fairness; leading indicators (contact rates) | Prometheus + Grafana; Evidently for drift reports; email alerts | Labels only arrive after each semester, so indirect signals must be monitored during the semester |
| **Deployment** | Packaging, testing, deployment | Docker; GitHub Actions (lint, test, build) | Consistent environments; automated tests before deployment |

## 4.4 Key Architectural Decisions

- **Batch scoring, no real-time API** (NFR1, Part 5): simple, cheap, sufficient for the weekly outreach rhythm; same-day absence calls continue under the existing process.
- **Monolithic pipeline:** a small data team (2–3 people), one model, evolving requirements; microservices bring no benefit yet.
- **Human-in-the-loop in two places:** a responsible person approves each new model; advisors and care staff decide whether to contact each student.
- **Tiering by contact cost:** the scarce, expensive resource (advisors) is reserved for the highest-risk group; the cheaper, existing resource (care team) covers the next group. The 10% and 25% cut-offs are configuration parameters, adjusted each semester to actual capacity.
- **Feedback loop:** interventions and next-semester outcomes return to the warehouse for evaluation and retraining. Intervened students must be flagged during training, because intervention changes the outcome (a student may stay *because* they were helped).
- **Fallbacks:** all model versions remain in the registry; bad data triggers the fallback rule; a failed scoring run keeps last week's list.

# 5. Trade-off Analysis

Part 5 · 15%

## 5.1 Accuracy vs. interpretability (simplicity vs. performance)

| | |
|---|---|
| **Conflict** | Gradient boosting usually ranks better than logistic regression but is harder to explain. Advisors need to know *why* a student is at risk to choose the right support; without clear reasons they will not trust the list. |
| **Options** | (a) Logistic regression only: easy to explain, possibly less accurate. (b) Gradient boosting with SHAP explanations: more accurate, indirect explanations, extra computation. |
| **Decision** | **Train both. Choose gradient boosting only if its time-based test PR-AUC exceeds logistic regression by at least 0.02; otherwise use logistic regression. In every case show the top 3 factors per student.** |
| **Revisit when** | Surveys show that advisors do not understand or trust the explanations. |

## 5.2 Data freshness vs. cost

| | |
|---|---|
| **Conflict** | Real-time scoring (on every absence or LMS login) gives the newest signal but needs streaming infrastructure, an always-on API and more operational work. |
| **Options** | (a) Real time. (b) Daily. (c) Weekly. |
| **Decision** | **Weekly. Source data only updates daily and advisors work on a weekly rhythm, so fresher scores would not change actions. Same-day reaction to absences is already handled by the care team's existing process. The cost is a job running a few minutes per week.** |
| **Revisit when** | The college introduces a service requiring same-day response (e.g. an urgent support line); then move to daily scoring, still not real time. |

## 5.3 Automation vs. human control

| | |
|---|---|
| **Conflict** | Automatically emailing warnings to every high-risk student would reach many students quickly without depending on staff time. But a model-based warning can be wrong, cause anxiety or a sense of being labelled, and push students further away. |
| **Options** | (a) Automatic warnings to students. (b) General support information sent to all students; personal contact by people in two tiers (advisors for Tier A, care team for Tier B). (c) Advisors only. |
| **Decision** | **Option (b). The system never tells a student "you are at risk of dropping out". Support information goes to everyone; personal contact is decided by advisors and the care team. Option (c) is rejected because it wastes the existing care-team capacity and exceeds advisor capacity. New models also require human approval.** |
| **Revisit when** | Capacity cannot keep up (7-day contact rate below 60% for several weeks); then narrow the tier cut-offs or add care staff before automating further. |

### 5.4 Privacy vs. accuracy

| | |
|---|---|
| **Conflict** | More detailed data (dormitory Wi-Fi logs, forum content, family or health information) might improve predictions but intrudes on privacy, may violate data protection rules and erodes student trust. |
| **Options** | (a) Use all collectable data. (b) Use only academic and financial data already collected for administration. |
| **Decision** | **Option (b), accepting lower accuracy. Sensitive attributes are not model inputs and are used only for fairness checks. Data is pseudonymised in the analytics store.** |
| **Revisit when** | The model fails the minimum PR-AUC; first engineer more features from academic data, and consider new data only with explicit student consent and legal review. |

### 5.5 Early detection vs. prediction accuracy

| | |
|---|---|
| **Conflict** | Earlier predictions leave more time to help but rely on less data (no interim results, few attendance records), so they are less accurate. Later predictions are more accurate but may be too late. |
| **Options** | (a) A single prediction at week 8. (b) Weekly scoring from week 3, accepting lower accuracy in early weeks. |
| **Decision** | **Option (b), with a "current week of semester" feature so the model knows how complete the data is. A separate target for week 5 (Recall@10% ≥ 35%); the dashboard clearly shows lower confidence in early weeks.** |
| **Revisit when** | Deployment data shows that interventions in weeks 3–5 are no more effective than at week 8. |

## 6. Risks, Limitations and Conclusion

| Risk | Mitigation |
|---|---|
| Labels only available after the semester; model quality cannot be measured during the semester | Monitor feature and score drift; full evaluation each semester when labels arrive |
| Feedback loop: successful interventions reduce dropout labels and bias the model | Log interventions; flag or exclude intervened cases when retraining; keep a control group |
| Changes in student behaviour (e.g. more online study, LMS change) | PSI > 0.2 triggers retraining; validation blocks schema changes |
| Students called twice by advisors and the care team, or missed between them | One contact owner per student per week (FR2); shared logging; monitor duplicate-call rate |
| Advisors or care staff do not use the system | Co-design with users; short lists with reasons; track usage as a system metric |
| Bias against a student group | Per-group recall in every version; fairness condition in the quality gate |
| Personal data leakage | Pseudonymisation, access control, audit logs, encryption, purpose limitation |

**Conclusion.** The system is built around a practical constraint: the **limited capacity of 40 advisors** who also teach, and a **student care team** that is ready to call students but currently follows a rigid rule. The model does not replace people; it helps both groups spend their time on the right students, about 8 weeks earlier than the

current mechanism, without duplicate contact. The value lies in the whole chain: validated data, honest time-based evaluation, human approval before deployment, explanations that support action, and monitoring that detects when the model no longer fits. The biggest risk is not the algorithm but whether support actually helps students stay, so measuring impact with a control group should be a priority from the first semester.

## References


1. Kästner, C. (2025). *Machine Learning in Production: From Models to Products*. MIT Press.
2. Hulten, G. (2018). *Building Intelligent Systems: A Guide to Machine Learning Engineering*. Apress.
3. Huyen, C. (2022). *Designing Machine Learning Systems*. O'Reilly Media.
4. Sculley, D. et al. (2015). Hidden Technical Debt in Machine Learning Systems. *NeurIPS*.
5. Bernardi, L. et al. (2019). 150 Successful Machine Learning Models: 6 Lessons Learned at Booking.com. *KDD*.
6. Government of Vietnam (2023). Decree 13/2023/ND-CP on Personal Data Protection.
7. DDM501 lecture notes: Lesson 1 (ML in Production), Lesson 2 (Requirements Engineering), Lesson 4 (Architecture and System Design).